Stories of Superbugs: Historical Narratives of Penicillin & Antimicrobial Resistance (AMR) at the London Science Museum

Brief Description of Work: In February 2025, I presented this work at the Triangle Undergraduate Literature Conference at Duke University. I included the corresponding presentation and poster made for the celebration of undergraduate research at UNC.

Link to the poster:
https://our.unc.edu/wp-content/uploads/sites/1148/gravity_forms/159-ad7f2de0736bb70ca3f153a194936e1c/2025/04/Vona-Poster-2025-7.pdf

Link to presentation: https://canva.link/i5pdspd6f114otx

Abstract:

Museum scholars argue that exhibits are not merely repositories of artifacts; they are narrative spaces that construct meaning and shape public understanding through storytelling. Thus, while museums may not be traditionally considered literature, analyzing their narrative characteristics demonstrates the impact of stories on society. This article analyzes Antimicrobial Resistance (AMR) narratives in the London Science Museum's Medicine Gallery, exploring how these stories suggest (and constrain) the solutions needed to ameliorate this global health crisis.

The exhibit's focus on penicillin as the first antibiotic reinforces a demarcated antimicrobial *era* and obscures AMR's endemic realities. This limits our imagination of the pre- and post-antimicrobial era to apocalyptic thinking. Furthermore, adherence to Goldworth et al's "quest narrative" retelling of the discovery of penicillin inflates Fleming's status and

serendipitous discovery, while minimizing the collaborative and iterative nature of scientific progress. By focusing on individual antibiotic misuse while neglecting systemic factors such as industrial agriculture and healthcare practices, the exhibit places the responsibility for mitigating AMR on consumers. Discussions of structural reform are further sidelined by an emphasis on technological solutions, reflecting the techno-optimistic bias in approaching global health crises.

By critically analyzing museums as narrative spaces, this article elucidates the role of public narratives in shaping our understanding of global health challenges. Recognizing the power of museum narratives on society is the first step to challenging oversimplified stories and advocating for nuanced, systemic approaches to addressing global health crises like AMR.

500-word sample:

The museum unduly emphasizes penicillin, elevating it as a defining symbol of the antimicrobial era and obscuring the complexities of antimicrobial history. Indeed, penicillin is the sole antimicrobial explicitly named or shown in this exhibit, and it is also aggrandized: vials of penicillin mold isolates and syringes used to give those early doses decorate the exhibit, the words "Penicillin was more valuable than Gold" are printed across the wall, "Life Changing Penicillin" punctuates a large title card in bold, red letters (Figure 1) (Figure 2). The combination of material objects and eye-catching proclamations of penicillin's importance quickly dominates the exhibit. The title cards further elevate the role of penicillin by stating: "As the first antibiotic, penicillin changed our lives. What began as a chance discovery…[became] an everyday way to fight bacterial infections… [and] sparked a hunt for new antibiotics" (Figure 3). Notably, these claims focus on penicillin itself, rather than antimicrobials as a whole. The importance of penicillin is undeniable, but this drug only represents one of over a hundred antimicrobials in this therapeutic modality and the resulting improvements to our lifespans.

The exhibit attempts to justify this subtle synecdochic framing of penicillin for antimicrobials by calling penicillin the “first antibiotic” and by arguing that it sparked the development of further antimicrobial agents. However, penicillin was not the first antimicrobial: In 1909, Ehrlich developed Salvarsan, the first “magic bullet” selective drug that targeted disease-causing microbes and left humans relatively unscathed. By 1935, the “miracle” sulfonamides, a whole class of antibiotics, had revolutionized treatment and sparked widespread attempts to develop more antimicrobials. Fleming published his initial paper on the effects of penicillin in a petri dish in 1929, so an argument could be made that penicillin was the second antimicrobial. However, he did not test its therapeutic effectiveness in vivo on an animal model. It was only over a decade later, in 1939, when Florey and Chain isolated the active ingredient and took the critical step of testing it on humans: but this was after the sulfonamides expanded the search for antimicrobials. Thus, the claim that penicillin “sparked the hunt” is misleading: efforts were already well underway. Penicillin ultimately did supplant the sulfa drugs due to an increased pharmaceutical base and greater impact on patient care- penicillin is a vital player in the story of antimicrobials- however, the continued use of the sulfa drugs to treat bacterial infections speaks to their utility and importance. Reducing the antimicrobial story to that of penicillin and misleadingly retelling the story clouds the nature of scientific progress and discovery. As society faces antimicrobial resistance, we must be open to a diverse range of paths forward to reclaim the efficacy of these lifesaving drugs, and not let a narrow story obscure our understanding of how we will get back to the golden age of therapeutics.

Addendum (Figures):

Figure 1: Vials of Penicillin

Figure 2: Wall of the Exhibit

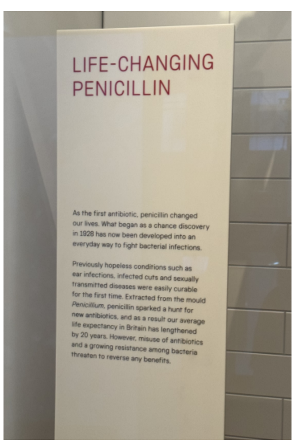

Figure 3: "Life-changing penicillin"